Measures of Central Tendency (2)

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Median

- Middle value of the observation.
- After they have been arranged in ascending or descending order.

Median for Ungrouped data

- There are two types of formula for ungrouped data

When "n" is odd:

$$Me = \left(\frac{n+1}{2}\right)^{th} Observation$$

When "n" is even:

$$Me = \frac{\left(\frac{n}{2}\right)^{th} Obs. + \left(\frac{n}{2}+1\right)^{th} Obs.}{2}$$

“n” is odd

- Step 1: Organize in ascending order
- Step 2: $Me = \left(\frac{n+1}{2}\right)^{th} Observation$
- For example: 6,8,4,7,2,9,3 are exam score of section “A”.

Organize in ascending order:

$2, 3, 4, 6, 7, 8, 9$

Here, $n = 7$ is an odd number. Then the median can be written as,

$$Me = \left(\frac{n+1}{2}\right)^{th} Obs.$$

$$\therefore Me = 6$$

Comment: The median score of section “A” is 6

“n” is even

- Step 1: Organize in ascending order
- Step 2: $Me = \frac{\left(\frac{n}{2}\right)^{th} Obs. + \left(\frac{n}{2}+1\right)^{th} Obs.}{2}$
- For example: 5,8,4,7,2,9 are exam score of section “A”.

Organize in ascending order:

$$2, 4, 5, 7, 8, 9$$

Here, $n = 6$ is an even number. Then the median can be written as,

$$Me = 6$$

Comment: The median score of section “A” is 6

Median for grouped data

$$Me = L_m + \frac{\frac{N}{2} - F_c}{f_m} \times c$$

$L_m = Lower\ limit\ of\ median\ class$

$N = Total\ number\ of\ observations$

$F_c = Cumultative\ frequency\ of\ pre - median\ class$

$f_m = Frequency\ of\ median\ class$

$c = Class\ interval$

Steps…

- Prepare a less than type cumulative frequency distribution.
- Determine $\frac{N}{2}$, where N is the total frequency.
- Locate the median class whose cumulative frequency includes the value of $\frac{N}{2}$.
- Determine the value of $L_m, F_c, f_m, and\ c$.

Median for grouped data

Class	Frequency	CF_i
2-6	2	2
6-10	6	8
10-14	4	12
14-18	2	14

Here, $N = 14$.

Now, $\frac{N}{2} = \frac{14}{2} = 7$

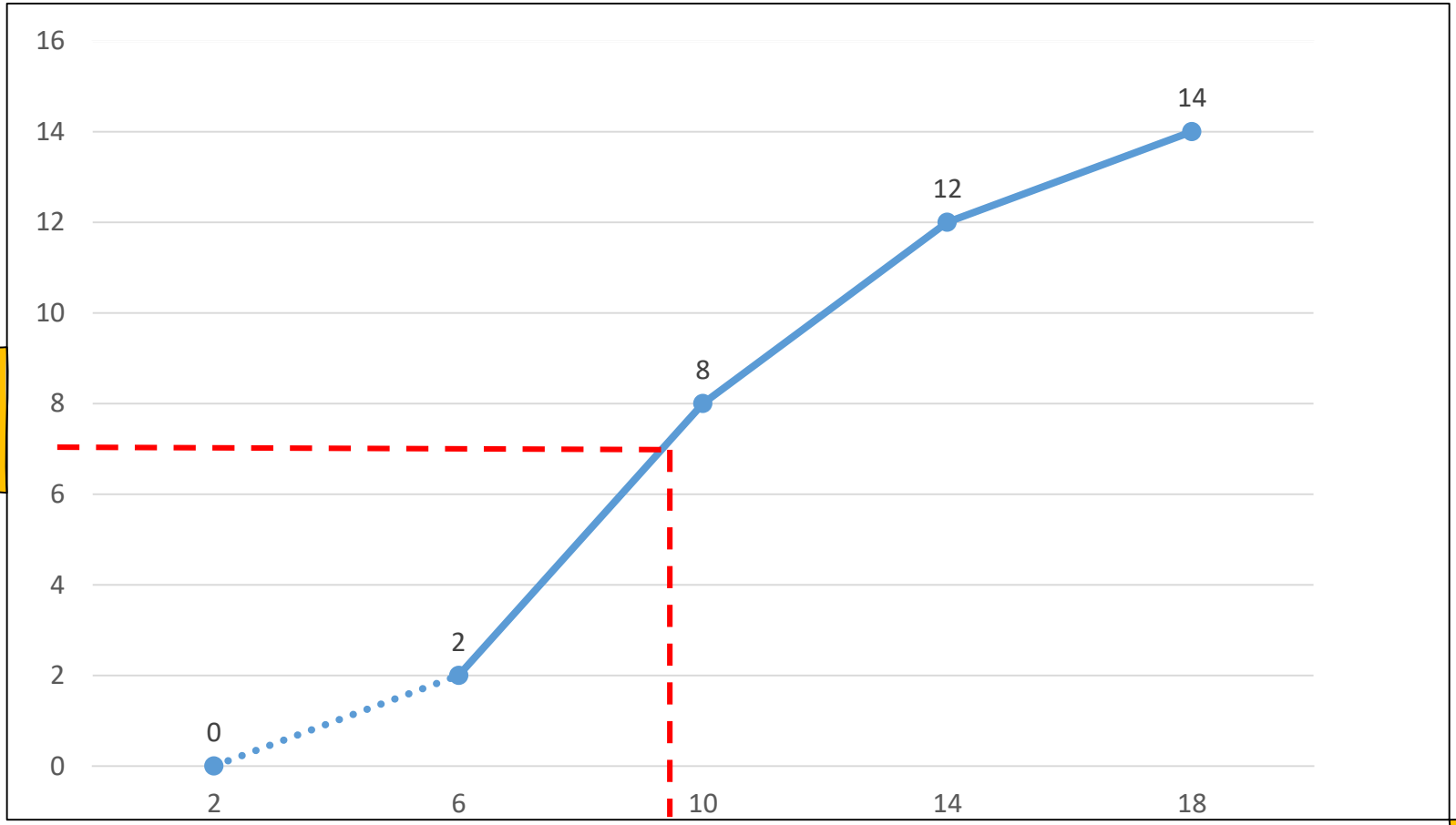
So, the median class is (6-10)

We know that,

$$Me = L_m + \frac{\frac{N}{2} - F_c}{f_m} \times c$$

$$\therefore Me = 6 + \frac{7 - 2}{6} \times 4 = 9.33$$

Median from graph



$\frac{N}{2} = 6.5$

$Me \approx 9.33$

Mode

- One mode = Unimodal
- Two mode = Bi-modal
- More than two = Multimodal

- Most frequent value
- Occurs more than one times.

- For example: 1, 2, 5, 5, 3, 6, 6, 6, 4,3

- Here the most frequent value is 6.
- Thus the mode value is 6.

Mode

- Example: 1, 2, 3, 4, 5, 6, 7, 8, 9, 10

- There is no value which occur more than one time. Thus there is no “Mode” in this data set.

Mode for grouped data

$$Mo = L_o + \frac{\Delta_1}{\Delta_1 + \Delta_2} \times c$$

L_o = Lower limit of modal class

Δ_1 = Difference between frequency of the modal class and pre-modal class.

Δ_2 = Difference between frequency of the modal class and post-modal class.

$c = Class\ interval$

Mode for grouped data

Class	f_i
2-6	2
6-10	6
10-14	4
14-18	2

Here, the class with highest frequency is (30-40). Thus, this class is our modal class.

Now, $Mo = L_o + \frac{\Delta_1}{\Delta_1+\Delta_2} \times c$

$\therefore Mo = 6 + \frac{4}{4+2} \times 4 = 8.67$

Some points…

- Does all data have a Mean, median, and mode?
- Every set of continuous data possesses a median, mode, and mean.
- When considering ordinal data, it encompasses solely a median and mode.
- Nominal data solely involves a mode

Measures of Location

- We have learned that median divided a set of data into two equal parts.
- In the same way, we can divides a set of data into

1. Four equal parts — Quartiles
2. Ten equal parts — Deciles
3. Hundred equal parts — Percentiles

Q, D, and P

- Quartiles: Divide the data set into four equal parts. So, we get 3 quartile points: Q1, Q2, and Q3

- Deciles: Divide the data set into ten equal parts. So, we get 9 decile points: D1, D2, D3, … , D9.

- Percentiles: Divide the data set into hundred equal parts. So, we get 99 percentile points: P1, P2, P3, … , P99

Q, D, and P (Ungrouped)

$$position\ of\ Q_i = \frac{i \times N}{4}$$

$i = 1, 2, 3$
$N = Total\ number\ of\ observations$

$$position\ of D_i = \frac{i \times N}{10}$$

$i = 1, 2, 3, 4, \ldots, 9$
$N = Total\ number\ of\ observations$

$$position\ of P_i = \frac{i \times N}{100}$$

$i = 1, 2, 3, 4, \ldots, 99$
$N = Total\ number\ of\ observations$

Steps…

$$position\ of\ D_i = \frac{i \times N}{10}$$

$$position\ of\ P_i = \frac{i \times N}{100}$$

- Arrange data set from smallest to largest
- Identify the position of Q_i by utilizing the formula,

$$J = \frac{(i \times n)}{4}$$

- If J is the integer value, then

$$\frac{\left(J^{th} Observation + (J+1)^{th} Observation\right)}{2}$$

- If J is not integer value, then take the next integer value as position.

Example (Ungrouped)

Data: 10, 12, 17, 13, 19

Organize the data into ascending order, 10, 12, 13, 17, 19

Now, position of $Q_1 = \frac{i \times N}{4} = 1.25$

Since, the position value is not integer, thus we should go for next integer value.

$$\therefore Q_1 = 2nd\ obs. = 12$$

Data: 10, 12, 17, 13

Organize the data into ascending order, 10, 12, 13, 17

Now, position of $Q_1 = \frac{i \times N}{4} = 1$

Since, the position value is integer

$$\therefore Q_1 = \frac{1st\ Obs. + 2nd\ Obs}{2} = 11$$

Q, D, and P (Grouped)

$$Q_i = L_Q + \frac{\frac{i \times N}{4} - F_c}{f_Q} \times c$$

$L_Q = Lower\ limit\ of\ quartile\ class$

$N = Total\ number\ of\ observations$

$F_c = Cumultative\ frequency\ of\ pre - quartile\ class$

$f_Q = Frequency\ of\ quartile\ class$

$c = Class\ interval$

Q, D, and P (Grouped)

$$D_i = L_D + \frac{\frac{i \times N}{10} - F_c}{f_D} \times c$$

$L_D = Lower\ limit\ of\ decile\ class$

$N = Total\ number\ of\ observations$

$F_c = Cumultative\ frequency\ of\ pre - decile\ class$

$f_c = Frequency\ of\ decile\ class$

$c = Class\ interval$

Q, D, and P (Grouped)

$$P_i = L_P + \frac{\frac{i \times N}{100} - F_c}{f_P} \times c$$

$L_P = Lower\ limit\ of\ percentile\ class$

$N = Total\ number\ of\ observations$

$F_c = Cumultative\ frequency\ of\ pre - percentile\ class$

$f_P = Frequency\ of\ percentile\ class$

$c = Class\ interval$

Self Practice

Age (in Years)	f_i
10-20	8
20-30	14
30-40	35
40-50	20
50-60	15
60-70	8

Determine $Median$

Determine $Q_1, Q_2, and\ Q_3$ with interpretation.

Determine $D_6, P_{36}, and\ P_{96}$ with interpretation.

Thank You!